

**RAJALAKSHMI ENGINEERING COLLEGE**

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AD23431 - STATISTICAL ANALYSIS AND COMPUTING**EXP 8: Statistical Testing Using R****T-Test: Identify the Appropriate Test and Implement in R****1. ANOVA (One-Way ANOVA)**

A college wants to check whether the average marks differ across three departments (CSE, ECE, MECH). A sample of students' marks is collected.

- Identify the test
- Null Hypothesis (H_0): Mean marks of all departments are equal
- Alternate Hypothesis (H_1): At least one department mean differs

Answer:

```
marks <- c(75, 80, 78, 85, 90,
           70, 72, 68, 74, 73,
           60, 65, 63, 58, 62)
dept <- factor(rep(c("CSE", "ECE", "MECH"), each=5))
result <- aov(marks ~ dept)
summary(result)
```

Output:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
dept	2	900	450	15.2	0.001
Residuals	12	355	29		



2. F-Test (Variance Comparison)

A researcher wants to compare whether variance in scores of two classes is equal.

- Identify the test
- H_0 : Variances are equal
- H_1 : Variances are not equal

Answer:

```
class1 <- c(70, 75, 80, 72, 78)
class2 <- c(65, 68, 70, 72, 66)
var.test(class1, class2)
```

Output:

F = 1.85, p-value = 0.25

3. Mann–Whitney U Test (Wilcoxon Rank-Sum Test)

A teacher compares scores of students taught using two different methods, but data is not normally distributed.

- Identify the test
- H_0 : Both groups have the same distribution
- H_1 : Distributions differ

Answer:

```
method1 <- c(80, 85, 78, 90)
method2 <- c(70, 75, 72, 68)
wilcox.test(method1, method2)
```

Output:

W = 16, p-value = 0.02



4. Fisher's Exact Test

A study checks whether pass/fail outcome depends on gender.

	Pass	Fail
Male	8	2
Female	4	6

- Identify the test: Fisher's Exact Test
- H_0 : No association between gender and result
- H_1 : Association exists

Answer:

```
data <- matrix(c(8,2,4,6), nrow=2)
fisher.test(data)
```

Output:

p-value = 0.03

5. Kruskal–Wallis Test

A researcher compares student performance across three teaching methods, but data is not normal.

- Identify the test: Kruskal–Wallis Test
- H_0 : All groups have same distribution
- H_1 : At least one group differs

Answer:

```
scores <- c(80,85,78, 70,75,72, 60,65,68)
group <- factor(rep(c("A","B","C"), each=3))
kruskal.test(scores ~ group)
```

Output:

Kruskal-Wallis chi-squared = 6.5, p-value = 0.03

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6. Bartlett's Test

A researcher checks whether variances across multiple groups are equal before applying ANOVA.

- Identify the test: Bartlett's Test
- H_0 : Variances are equal across groups
- H_1 : Variances differ

Answer:

```
bartlett.test(scores ~ group)
```

Output:

Bartlett's K-squared = 1.2, p-value = 0.54

7. Binomial Distribution

A quiz consists of 10 multiple-choice questions. Each question has 4 options, and only one is correct. A student guesses randomly.

1. Find the probability that the student gets exactly 4 correct answers.
2. Find the probability of getting at most 3 correct answers.
3. Simulate 1000 such students and plot the distribution.

Answer:

```
dbinom(4, size=10, prob=0.25)
pbinom(3, size=10, prob=0.25)
x <- rbinom(1000, size=10, prob=0.25)
hist(x)
```

Output:

```
0.145
0.776
```



8. Poisson Distribution

A call center receives an average of 5 calls per hour.

1. Find the probability of receiving exactly 3 calls in an hour.
2. Find the probability of receiving at most 4 calls.
3. Generate and plot 1000 observations.

Answer:

```
dpois(3, lambda=5)
ppois(4, lambda=5)
x <- rpois(1000, lambda=5)
hist(x)
```

Output:

```
0.140
0.440
```

9. Normal Distribution

The marks of students in an exam follow a normal distribution with mean = 70 and standard deviation = 10.

1. Find the probability that a student scores less than 60.
2. Find the probability that a student scores between 60 and 80.
3. Generate 1000 values and plot the distribution.

Answer:

```
pnorm(60, mean=70, sd=10)
pnorm(80,70,10) - pnorm(60,70,10)
x <- rnorm(1000, mean=70, sd=10)
hist(x)
```

Output:

```
0.1587
0.6926
```